POSTCOMPOSITION WITH AN ETALE MORPHISM PRESERVES AT-WORST-NODAL MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the smooth-point case of the fibre statement is the standard fact that smoothness is stable under field extension (the source states it as 'the analogous result for smooth points' without a reference); the nodal case is Lemma lemma-node-over-separable-extension (in-source, navigation stub); the flatness and finite presentation steps cite the standard Morphisms lemmas (external tooling).

Editorial note. The label lemma-nodal-family-postcompose-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Fibral definition of at-worst-nodal morphisms. Same unit, the definition of at-worst-nodal morphisms of relative dimension 1. The proof reduces to the fibres because the at-worst-nodal condition on a flat locally finitely presented morphism is a condition on its fibres, by definition.
- Nodal and smooth points under separable extension. Same unit, Lemma lemma-node-over-separable-extension. The nodal case of the fibre statement follows from Lemma lemma-node-over-separable-extension; the smooth-point case is the standard stability of smoothness under field extension.

Source reference. Original reference 0CD6: sources/stacks-curves/pinned-original-source.tex, source lines 4684–4684. Prerequisite pointer invoked at proof line(s) 5035; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01TR: sources/stacks-chow/tags.txt, source lines 2384–2384. Prerequisite pointer invoked at proof line(s) 5031; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01U7: sources/stacks-chow/tags.txt, source lines 2400–2400. Prerequisite pointer invoked at proof line(s) 5032; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02GS: sources/stacks-chow/tags.txt, source lines 3239–3239. Prerequisite pointer invoked at proof line(s) 5030; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02GR: sources/stacks-chow/tags.txt, source lines 3238–3238. Prerequisite pointer invoked at proof line(s) 5029; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $S' \to S$ be an étale morphism of schemes. Let $f : X \to S'$ be at-worst-nodal of relative dimension 1. Then the composition $g : X \to S$ is at-worst-nodal of relative dimension 1.*

Proof. Observe that g is flat and locally of finite presentation as a composition of morphisms which are flat and locally of finite presentation (use Morphisms, Lemmas 02GR, 02GS, 01TR, and 01U7). Thus it suffices to prove the fibres of g have at-worst-nodal singularities. This follows from Lemma 0CD6 and the analogous result for smooth points. □